

On periodic subgroups of factorized groups

By

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1. Introduction. Let the group $G = AB$ be the product of two subgroups A and B . If N is a normal subgroup of G , the *factorizer* of N in G is the subgroup $X(N) = AN \cap BN$. It is easy to show that $X(N)$ has the triple factorization

$$X(N) = (A \cap BN)(B \cap AN) = (A \cap BN)N = (B \cap AN)N.$$

Thus the investigation concerning factorized groups very often leads to the consideration of groups with a triple factorization of the form

$$G = AB = AK = BK,$$

where K is a normal subgroup of G . It has recently been proved by Robinson and Stonehewer [6] that if the factors A , B and K are abelian, then the group G has a central series. However, an example of Sysak [7] shows that such a group G is not always locally nilpotent, even if K has rank 1.

The aim of this article is to study the behaviour of elements of finite order in groups with a triple factorization by locally nilpotent subgroups. An arbitrary periodic locally soluble group $G = AB = AK = BK$, factorized by three locally nilpotent subgroups A , B and K , with K normal in G , need not be locally nilpotent, since Sysak [7] has constructed a group of this type, where K is an infinite minimal normal subgroup of G . On the other hand Amberg [1] proved that G is locally nilpotent, provided that A and B are hypercentral. Improving this theorem, we shall prove that every periodic group $G = AB = AK = BK$ factorized by three locally nilpotent subgroups A , B and K , with K normal in G , is locally nilpotent if one of the factors A and B is hyperabelian. This will be a consequence of results on groups having a triple factorization by subgroups with normal Sylow π -subgroups for a fixed set π of primes.

In the general case, information are obtained on the structure of periodic subgroups of groups with a locally nilpotent triple factorization. In fact, it will be proved that if the hyperabelian group $G = AB = AK = BK$ is the product of two locally nilpotent subgroups A and B and a nilpotent normal subgroup K , then every periodic normal subgroup of G is locally nilpotent. This can be used to show that the set of all elements of finite order of G is a locally nilpotent subgroup, provided that either G is locally polycyclic or G is soluble and one of the factors A and B has finite torsion-free rank.

Most of our notation is standard and can be found in [5]. In particular:

$\pi(G)$ is the set of prime divisors of orders of elements of G .

$O_\pi(G)$ is the maximum periodic normal π -subgroup of G , for a set π of primes.

π' is the complementary set of primes to π .

A group G has *finite torsion-free rank* if it has a series of finite length whose factors are either periodic or infinite cyclic.

A subgroup H of a factorized group $G = AB$ is called *factorized* if $A \cap B \leq H$ and $H = (A \cap H)(B \cap H)$. Thus H is a factorized subgroup if and only if $X(H) = H$.

2. Statements and proofs. Let π be a set of primes. A group G is called π -closed if the set of all π -elements of G is a subgroup. In particular, if p is a prime, a group G is p -closed if and only if it has a unique Sylow p -subgroup. Thus a locally finite group is locally nilpotent if and only if it is p -closed for each prime p . If the group G contains a π -closed normal subgroup N such that G/N has no non-trivial π -elements, it is clear that G is π -closed. Moreover, if the group G is π -closed and N is a periodic normal subgroup of G , the factor group G/N is also π -closed.

In order to study products of π -closed groups, the following result of Zaicev [8] is needed (see also [4], Lemma 4.6.4).

Lemma 1. *Let the group $G = AB$ be the product of two subgroups A and B , and let A_0 and B_0 be normal subgroups of A and B , respectively. If A locally satisfies the maximal condition on subgroups and $x = ab$ ($a \in A$, $b \in B$) is an element of G normalizing the subgroup $H = \langle A_0, B_0 \rangle$, then a normalizes $Z(H) \cap A_0$.*

Lemma 2. *Let π be a set of primes, and let the group $G = AB = AK = BK$ be the product of two π -closed subgroups A and B , and a normal subgroup K such that $A \cap K$ and $B \cap K$ are periodic and $\pi(K) \cap \pi$ is empty. If A locally satisfies the maximal condition on subgroups and A_0 is a periodic abelian normal π -subgroup of A such that A_0^G is locally finite, then A_0 is normal in G and is contained in B .*

Proof. As B is π -closed, the set B_1 of all π -elements of B is a normal subgroup of B . Then $B_1 K$ is normal in $G = BK$, and the factor group $G/B_1 K \cong B/B_1 (B \cap K)$ has no π -elements, so that A_0 is contained in $B_1 K$. Since $A_0 K$ is normal in $G = AK$, the subgroup $B_0 = A_0 K \cap B_1$ is normal in B . Moreover $A_0 K = A_0 K \cap B_1 K = B_0 K$, with $A_0 \cap K = B_0 \cap K = 1$, and hence $B_0 \cong A_0$ is abelian. The subgroup $H = \langle A_0, B_0 \rangle$ is contained in $A_0 K = B_0 K$, and so $H = A_0(H \cap K) = B_0(H \cap K)$, where $H \cap K$ is normal in H .

Let a_0 be an element of A_0 of order a power of a prime q , and write $a_0 = b_0 h$, with b_0 in B_0 and h in $H \cap K$. Then $\langle a_0 \rangle$ and $\langle b_0 \rangle$ are Sylow q -subgroups of $\langle a_0 \rangle (H \cap K) = \langle b_0 \rangle (H \cap K)$, and so also of $\langle a_0, b_0 \rangle$. As $\pi(K) \cap \pi = \emptyset$, the factor group $K A_0^G / A_0^G$ has no π -elements, so that a_0 and b_0 both belong to A_0^G . It follows that $\langle a_0, b_0 \rangle$ is a finite group, and hence there exists an element x of $H \cap K$ such that a_0^x lies in $\langle b_0 \rangle \leq B_0$. Let $x = ab$, with a in A and b in B . Then $a_0^x = a_0^{xb^{-1}}$ is an element of $B_0^{b^{-1}} = B_0$, so that a_0^a belongs to $A_0 \cap B_0 \leq Z(H)$. In particular a_0^a is in $Z(H) \cap A_0$. On the other hand, a

normalizes $Z(H) \cap A_0$ by Lemma 1, so that a_0 belongs to $Z(H)$, and hence $a_0 = a_0^x$ is in B_0 . Therefore A_0 is contained in $B_0 \leq B$, and so $A_0 = B_0$ is normal in $G = AB$. $\square$

We can now prove the following result on groups with a triple factorization by π -closed subgroups.

Theorem 3. *Let π be a set of primes, and let the group $G = AB = AK = BK$ be the product of three π -closed subgroups A , B and K , where K is a locally finite normal subgroup of G . If one of the factors A and B is hyperabelian and locally polycyclic, then G is π -closed.*

Proof. It is clearly enough to show that the factor group $G/O_\pi(G)$ is π -closed, so that we may suppose $O_\pi(G) = 1$. In particular the π -closed locally finite normal subgroup K of G is a π' -group. Assume that A is hyperabelian and locally polycyclic. It follows from Lemma 2 that A has no non-trivial abelian normal π -subgroups, so that $\pi(A) \cap \pi = \emptyset$, as A is π -closed. Then also the factor group $G/K \cong A/(A \cap K)$ has no π -elements, so that $\pi(G) \cap \pi = \emptyset$, and G is π -closed. $\square$

The previous theorem has the following consequences.

Corollary 4. *Let π be a set of primes, and let the locally finite group $G = AB = AK = BK$ be the product of three π -closed subgroups A , B and K , where K is normal in G . If one of the factors A and B is hyperabelian, then G is π -closed.*

Corollary 5. *Let the group $G = AB = AK = BK$ be the product of three locally nilpotent subgroups A , B and K , where K is a periodic normal subgroup of G . If one of the factors A and B is hyperabelian, then the set of all elements of finite order of G is a locally nilpotent subgroup. In particular, if G is periodic, it is locally nilpotent.*

Corollary 5 can be used to obtain the following property of metanilpotent products of periodic locally nilpotent groups. This was already known in the case of a product of two periodic abelian groups (see [2]).

Corollary 6. *Let the metanilpotent group $G = AB$ be the product of two periodic locally nilpotent subgroups A and B . Then the Hirsch-Plotkin radical of G is a factorized subgroup.*

Proof. It is well-known that the group G is periodic (see [4], Theorem 3.2.6). Let H be the Hirsch-Plotkin radical of G . The factorizer $X = X(H)$ of H in G has the triple factorization

$$X = A^* B^* = A^* H = B^* H,$$

where $A^* = A \cap BH$ and $B^* = B \cap AH$, and hence it is locally nilpotent by Corollary 5. As the factor group G/H is nilpotent, the subgroup X is subnormal in G , so that $X \leq H$. Thus $X = H$, and H is a factorized subgroup of G . $\square$

In the above results the assumption that the normal subgroup K is periodic was used to produce large periodic normal subgroups. Now we investigate the behaviour of

elements of finite order in a triply factorized group $G = AB = AK = BK$, when the normal subgroup K is not periodic.

Proposition 7. *Let the hyperabelian group $G = AB = AK = BK$ be the product of two locally nilpotent subgroups A and B , and a nilpotent normal subgroup K . Then every periodic normal subgroup of G is locally nilpotent.*

Proof. Let T be the maximum periodic normal subgroup of G . If the result holds for the factor group G/K' , then TK'/K' is locally nilpotent, so that TK/K' is locally nilpotent, and hence also $T \leq TK$ is locally nilpotent. Thus we may suppose that K is abelian. As locally finite residually locally nilpotent groups are locally nilpotent, it is enough to show that $T/O_{p'}(G)$ is locally nilpotent for each prime p . Hence without loss of generality it can be assumed that $O_{p'}(G) = 1$ for some prime p , so that in particular $\pi(K) \subseteq \{p\}$.

Suppose first that $A \cap K$ and $B \cap K$ both are periodic. Let A_0 be a periodic abelian normal p' -subgroup of A which is contained in TK . The normal subgroup TK of G is periodic-by-abelian, so that T is the set of all elements of finite order of TK . Thus $A_0 \leq T$, and so $A_0^G \leq T$ is locally finite. Application of Lemma 2 yields that A_0 is normal in G , and hence $A_0 = 1$. Therefore $A_1 = A \cap TK$ does not contain periodic abelian non-trivial A -invariant p' -subgroups. As $A_1 \cap K$ and $A_1/(A_1 \cap K) \cong A_1 K/K$ are periodic, it follows that also the normal subgroup A_1 of A is periodic, so that A_1 is a p -group. Moreover $TK = AK \cap TK = A_1 K$, and hence $T/(T \cap K) \cong TK/K = A_1 K/K$ is a p -group. Therefore T is a p -group, and so it is locally nilpotent.

In the general case, the subgroups $A \cap K$ and $B \cap K$ both are normal in $G = AK = BK$, so that also $N = (A \cap K)(B \cap K)$ is a normal subgroup of G . It follows from the first part of the proof that the maximum periodic normal subgroup T_0/N of the factor group G/N is locally nilpotent. Let E be a finite subgroup of T . There exists a finitely generated subgroup E_1 of A such that $E \leq E_1 K$. Then $L = \langle E \cap A \cap K, E_1 \rangle$ is a finitely generated subgroup of A , and hence is nilpotent, so that $E \cap A \cap K \leq Z_n(L)$ for some positive integer n . As K is abelian, it follows that $E \cap A \cap K \leq Z_n(LK)$, and thus $E \cap A \cap K \leq Z_n(E)$. If $G^* = G/(A \cap K)$, the same argument shows that $E^* \cap B^* \cap K^* \leq Z_m(E^*)$ for some positive integer m . On the other hand, $E^*/(E^* \cap B^* \cap K^*) \cong E^*(B^* \cap K^*)/(B^* \cap K^*)$ is isomorphic to the finite subgroup EN/N of T_0/N , so that $E^*/(E^* \cap B^* \cap K^*)$ is nilpotent. Therefore $E^* = E(A \cap K)/(A \cap K) \cong E/(E \cap A \cap K)$ is nilpotent, and hence also E is nilpotent. This proves that T is locally nilpotent. $\square$

Our last two results show that in certain cases the set of all elements of finite order of the triply factorized group G is a subgroup.

Theorem 8. *Let the soluble group $G = AB = AK = BK$ be the product of two locally nilpotent subgroups A and B , and a nilpotent normal subgroup K . If one of the factors A and B has finite torsion-free rank, then the set of all elements of finite order of G is a locally nilpotent subgroup.*

Proof. Assume that A has finite torsion-free rank, and let T be the subgroup generated by all elements of finite order of G . If TK'/K' is periodic, it follows from Proposition 7 that TK'/K' is locally nilpotent. Thus TK/K' is locally nilpotent, so that TK is locally nilpotent, and hence the subgroup T is periodic and locally nilpotent. Therefore it is enough to show that the set of all elements of finite order of the factor group G/K' is a subgroup, and in particular we may suppose that K is abelian. Then $B \cap K$ is a normal subgroup of $G = BK$, and $B/(B \cap K) \cong A/(A \cap K)$ has finite torsion-free rank. Put $\bar{G} = G/(B \cap K)$. The locally nilpotent group A is periodic-by-nilpotent (see [5] Part 2, Theorem 6.36), and hence it has a series of finite length with central or periodic factors. Then it follows from Theorem 4.6.1 of [4] that the factorized group $\bar{G} = \bar{A}\bar{B}$ has finite torsion-free rank. If $\bar{V}$ is the maximum periodic normal subgroup of $\bar{G}$, the factor group $\bar{G}/\bar{V}$ has finite Prüfer rank (see [5] Part 2, Lemma 9.34). Then $\bar{G}/\bar{V}$ is locally nilpotent by Theorem B of [3], and hence is torsion-free, so that the subgroup $\bar{V}$ is the set of all elements of finite order of $\bar{G}$. Let E be the subgroup generated by a finite set of elements of finite order of G , and let E_1 be a finitely generated subgroup of B such that $E \leq E_1 K$. As $E/(E \cap B \cap K) \cong E(B \cap K)/(B \cap K)$ is finite, the subgroup $E \cap B \cap K$ is finitely generated, and hence $L = \langle E \cap B \cap K, E_1 \rangle$ is a finitely generated subgroup of B . Thus L is nilpotent and $E \cap B \cap K \leq Z_n(L)$ for some positive integer n . Since K is abelian, we have also that $E \cap B \cap K$ is contained in $Z_n(LK)$, and so $E \cap B \cap K \leq Z_n(E)$. Therefore $E/Z_n(E)$ is finite, and E is finite-by-nilpotent (see [5] Part 1, Corollary 2 to Theorem 4.21). Then E is finite, and the set of all elements of finite order of G is a subgroup. $\square$

Theorem 9. *Let the hyperabelian group $G = AB = AK = BK$ be the product of two locally nilpotent subgroups A and B , and a nilpotent normal subgroup K . If G is locally polycyclic, then the set of all elements of finite order G is a locally nilpotent subgroup.*

Proof. As in the proof of Theorem 8 we may suppose that the subgroup K is abelian, and it is enough to show that the set of all elements of finite order of G is a subgroup.

Suppose first that $A \cap K = B \cap K = 1$, and without loss of generality assume that the abelian normal subgroup K of G is torsion-free. Let A_0 be an abelian normal p -subgroup of A , where p is a prime, and suppose that A_0 is not contained in B . Let a be an element of $A_0 \setminus B$, and write $a = bx$, with b in B and x in $K \setminus \{1\}$. Let q be a prime such that $q \neq p$, and put $L = x^G$ and $M = (x^q)^G$. Assume that x belongs to M . Then there exists a finitely generated subgroup H of G such that x is in $V = (x^q)^H$. Hence x^q belongs to V^q , and so $V^q = V$. The subgroup $\langle x^q, H \rangle$ is finitely generated, and hence polycyclic, so that $V \leq \langle x^q, H \rangle \cap K$ is a finitely generated torsion-free abelian group, a contradiction since $V^q = V$. Therefore x is not in M , and L/M is a non-trivial abelian group of exponent q . Write $\bar{G} = G/M$, and let $\bar{X}$ be the factorizer of $\bar{L} = L/M$ in the factorized group $\bar{G} = \bar{A}\bar{B}$. Then

$$\bar{X} = \bar{A}^* \bar{B}^* = \bar{A}^* \bar{L} = \bar{B}^* \bar{L},$$

where $\bar{A}^* = \bar{A} \cap \bar{B} \bar{L}$ and $\bar{B}^* = \bar{B} \cap \bar{A} \bar{L}$. Clearly the set of all elements of finite order of $\bar{X}$ is a subgroup, so that in particular the normal closure $(\bar{A}_0 \cap \bar{A}^*)^{\bar{X}}$ is periodic. Application of Lemma 2 yields that $\bar{A}_0 \cap \bar{A}^*$ is contained in $\bar{B}^*$, so that $\bar{a}$ belongs to $\bar{B}^*$. As $\bar{b}$ also is in $\bar{B}^*$, it follows that $\bar{x}$ is an element of $\bar{B}^*$, so that $\bar{x}$ is in $\bar{B}$. Therefore x belongs

to $BM \cap L = M(B \cap L) = M$. This contradiction shows that A_0 is contained in B . The subgroup $A_0 K$ is normal in $G = AK$, so that $A_0 K \cap B = A_0(K \cap B) = A_0$ is normal in B . Thus A_0 is a normal subgroup of $G = AB$. It follows that every periodic abelian normal subgroup of A is normal in G . The subgroup T of all elements of finite order of A has an ascending A -invariant series with abelian factors, and the above argument proves that T is a normal subgroup of G . As $A \cap K = 1$, the factor group G/T is torsion-free, and hence T is the set of all elements of finite order of G .

In the general case, the subgroups $A \cap K$ and $B \cap K$ both are normal in $G = AK = BK$, so that $N = (A \cap K)(B \cap K)$ also is a normal subgroup of G . It follows from the first part of the proof that the theorem holds for the factor group G/N . Let E be the subgroup generated by a finite set of elements of finite order of G , and let E_1 be a finitely generated subgroup of A such that $E \leq E_1 K$. As E is polycyclic, the intersection $E \cap A \cap K$ is finitely generated, and hence $L = \langle E \cap A \cap K, E_1 \rangle$ is a finitely generated subgroup of A . Thus L is nilpotent, and $E \cap A \cap K \leq Z_n(L)$ for some positive integer n . As K is abelian, it follows that $E \cap A \cap K \leq Z_n(LK)$, and so $E \cap A \cap K \leq Z_n(E)$. Now the last part of the proof of Proposition 7 can be used to show that E is finite-by-nilpotent. Therefore E is finite, and hence the set of all elements of finite order of G is a subgroup. $\square$

References

- [1] B. AMBERG, Lokal endlich-auflösbare Produkte von zwei hyperzentralen Gruppen. Arch. Math. **35**, 228–238 (1980).
- [2] B. AMBERG, Products of two abelian subgroups. Rocky Mountain J. Math. **14**, 541–547 (1984).
- [3] B. AMBERG, S. FRANCIOSI and F. DE GIOVANNI, Groups with a nilpotent triple factorisation. Bull. Austral. Math. Soc. **37**, 69–79 (1988).
- [4] B. AMBERG, S. FRANCIOSI and F. DE GIOVANNI, Products of Groups. Oxford 1992.
- [5] D. J. S. ROBINSON, Finiteness conditions and generalized soluble groups. Berlin-Heidelberg-New York 1972.
- [6] D. J. S. ROBINSON and S. E. STONEHEWER, Triply factorizations by abelian groups. Arch. Math. **60**, 223–232 (1993).
- [7] Y. P. SYSAK, Products of infinite groups (Russian). Akad. Nauk Ukrain. SSR Inst. Mat. Kiev 1982.
- [8] D. I. ZAICEV, Soluble factorized groups (Russian). Structure of Groups and Subgroups Characterizations, 15–33, Akad. Nauk Ukrain. Inst. Mat. Kiev 1984.

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